

Mutual linearity from classical Markov response theory

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Mutual linearity of stationary probabilities and currents in Markov jump processes follows directly from classical Markov response theory. Using the exact finite-response identity expressed through the Drazin inverse, we show that changing the rates of one edge changes all stationary probabilities by the same scalar factor multiplied by fixed coefficients. Eliminating this factor gives the known mutual-linearity relations, valid arbitrarily far from equilibrium.

I. INTRODUCTION

Stationary probabilities and currents generally depend nonlinearly on transition rates. Nevertheless, varying the rates of one edge produces affine relations between these quantities, known as mutual linearity [1–6]. Our goal is to give a short common derivation from the classical finite-response identity of Markov-chain theory [7, 8]. Written with the Drazin inverse, this identity shows that all probability changes share one scalar factor. Eliminating it yields the known relations and makes their extension beyond Markov processes transparent.

II. EXACT STATIONARY RESPONSE

The finite-response identity below dates back to the 1960s [7]. We simply recast it here for Markov jump processes.

We consider a finite irreducible Markov jump process, $\partial_t \mathbf{p} = W \mathbf{p}$, where W_{ij} is the transition rate from j to i , and $W_{jj} = -\sum_{i \neq j} W_{ij}$. The stationary distribution satisfies $W \boldsymbol{\pi} = 0$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$. For a second irreducible generator $W' = W + \Delta W$, the classical fundamental-matrix identity takes the form

$$\Delta \boldsymbol{\pi} \equiv \boldsymbol{\pi}' - \boldsymbol{\pi} = -G \Delta W \boldsymbol{\pi}', \quad (1)$$

where $G = W^D$ is the Drazin inverse [9–11]. It satisfies $GW = WG = I - \boldsymbol{\pi} \mathbf{1}^\top$, $G \boldsymbol{\pi} = 0$, and $\mathbf{1}^\top G = 0$. Indeed, subtracting the two stationary equations gives $W \Delta \boldsymbol{\pi} = -\Delta W \boldsymbol{\pi}'$. Multiplication by G and normalization yield Eq. (1), without assuming that ΔW is small.

III. MUTUAL LINEARITY

A. Probabilities and observables

We change W_{ij} and W_{ji} , keeping all other off-diagonal rates fixed and adjusting the diagonal entries to preserve probability conservation. Only components i and j of $\Delta W \boldsymbol{\pi}'$ are

nonzero, and they have opposite signs. Equation (1) therefore gives

$$\begin{aligned} \Delta \pi_n &= (G_{nj} - G_{ni})(\Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i) \\ &\equiv h_n \alpha. \end{aligned} \quad (2)$$

Here $h_n \equiv G_{nj} - G_{ni}$ is calculated from the rates before the change, including W_{ij} and W_{ji} . We keep this initial W fixed and vary W'_{ij} and W'_{ji} ; all dependence on these new rates enters $\alpha \equiv \Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i$. Repeating Eq. (2) for state m and eliminating α gives

$$\Delta \pi_n = \frac{h_n}{h_m} \Delta \pi_m, \quad h_m \neq 0. \quad (3)$$

Thus (π'_m, π'_n) lies on one straight line for every choice of the new edge rates [4]. The coefficients h_n depend on the initial values of W_{ij} and W_{ji} , but their ratios, and hence this line, do not.

For state observables $O = \sum_n O_n \pi_n$ and $Q = \sum_n Q_n \pi_n$ with fixed weights, the same argument gives

$$\Delta O = \frac{\sum_n O_n h_n}{\sum_n Q_n h_n} \Delta Q, \quad (4)$$

provided the denominator is nonzero. Taking infinitesimal changes in Eq. (2) further yields

$$\partial_{W_{ij}} \pi_n = h_n \pi_j, \quad \partial_{W_{ji}} \pi_n = -h_n \pi_i, \quad (5)$$

so that $-\partial_{W_{ij}} \pi_n / \partial_{W_{ji}} \pi_n = \pi_j / \pi_i$ whenever $h_n \neq 0$, recovering the response ratio of Ref. [4]. Our dynamical fluctuation-response theory gives the same ratio for responses to impulsive rate perturbations at time t , with π_j / π_i replaced by $p_j(t) / p_i(t)$ [12]. This relation does not require mutual linearity under finite rate changes.

B. Currents

The stationary current from l to k is $J_{kl} = W_{kl} \pi_l - W_{lk} \pi_k$. For an edge whose transition rates W_{kl} and W_{lk} are kept fixed, Eq. (2) gives

$$\Delta J_{kl} = (W_{kl} h_l - W_{lk} h_k) \alpha. \quad (6)$$

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On the perturbed edge, the rates themselves also change:

$$\begin{aligned}\Delta J_{ij} &= W_{ij}\Delta\pi_j - W_{ji}\Delta\pi_i + \Delta W_{ij}\pi'_j - \Delta W_{ji}\pi'_i \\ &= (1 + W_{ij}h_j - W_{ji}h_i)\alpha.\end{aligned}\quad (7)$$

Eliminating α between the currents on (i, j) and another edge (k, l) gives

$$\Delta J_{kl} = \frac{W_{kl}h_l - W_{lk}h_k}{1 + W_{ij}h_j - W_{ji}h_i} \Delta J_{ij}, \quad (8)$$

which is the mutual linearity of stationary currents [1–3]. Both current changes contain the same α , so their ratio is independent of the new edge rates, provided the denominator is nonzero and the perturbed process remains irreducible.

The connection with the first-passage formulation of Ref. [2] follows from $h_n = \pi_n(\tau_{j \rightarrow n} - \tau_{i \rightarrow n})$, where $\tau_{j \rightarrow n}$ is the mean first-passage time from j to n and $\tau_{n \rightarrow n} = 0$ [9].

IV. DISCUSSION AND CONCLUSIONS

The argument applies to any linear system $A\mathbf{x} = \mathbf{b}$ with nonsingular A and fixed $\mathbf{b}$. If only A_{ij} changes and A' is non-

singular, subtracting the equations and eliminating $\Delta A_{ij}x'_j$ between n and m gives

$$\begin{aligned}\Delta x_n &= -(A^{-1})_{ni}\Delta A_{ij}x'_j, \\ \Delta x_n &= \frac{(A^{-1})_{ni}}{(A^{-1})_{mi}} \Delta x_m, \quad (A^{-1})_{mi} \neq 0.\end{aligned}\quad (9)$$

The same proportionality holds for fixed linear observables.

The same argument applies whenever $\Delta A\mathbf{x}' = \mathbf{u}\alpha$, with fixed $\mathbf{u}$: the perturbation enters the equations in fixed proportions, with only its scalar amplitude α changing. Then $\Delta\mathbf{x} = -A^{-1}\mathbf{u}\alpha$, so eliminating α again gives mutual linearity.

The frequency-domain results follow by setting $A = sI - W$. For time-independent rates, the Laplace-transformed master equation is $(sI - W)\tilde{\mathbf{p}}(s) = \mathbf{p}(0)$. Keeping the same initial distribution in both systems, define $R(s) = (sI - W)^{-1}$ for $\text{Re } s > 0$ and $h_n(s) = R_{ni}(s) - R_{nj}(s)$. Changing the rates of edge (i, j) and eliminating the common factor between states n and m gives

$$\begin{aligned}\Delta\tilde{p}_n &= h_n(s)(\Delta W_{ij}\tilde{p}'_j - \Delta W_{ji}\tilde{p}'_i), \\ \Delta\tilde{p}_n &= \frac{h_n(s)}{h_m(s)} \Delta\tilde{p}_m, \quad h_m(s) \neq 0.\end{aligned}\quad (10)$$

All transformed probabilities are evaluated at the same s . This recovers the Laplace-domain probability relations [5, 6]. Repeating Eqs. (6)–(8) with $\tilde{\mathbf{p}}(s)$ and $h_n(s)$ gives the corresponding current relations [1]. The frequency-dependent slope corresponds to a convolution in time, rather than an affine relation at each instant.

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